THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Balanced forces acting on a body: 1

(a) produce acceleration **(b)** always change its direction
(c) always change its speed **(d)** do not change its state of motion

2. Momentum of a body of mass m moving with velocity v is: 1

(a) mv squared **(b)** m + v **(c)** mv **(d)** m/v

3. The weight of a body is: 1

(a) its density **(b)** its mass
(c) the gravitational force on it **(d)** its volume

4. A body moving with uniform velocity has a net force of: 1

(a) equal to mass **(b)** maximum
(c) zero **(d)** equal to its weight

5. The SI unit of momentum is: 1

(a) J **(b)** N **(c)** kg m/s squared **(d)** kg m/s

6. The weight of a body of mass 5 kg is (g = 10 m/s squared): 1

(a) 5 N **(b)** 0.5 N **(c)** 500 N **(d)** 50 N

7. A heavy and a light body are pushed with the same force. The lighter one has: 1

(a) greater acceleration **(b)** smaller acceleration
(c) no acceleration **(d)** the same acceleration

8. A force of 10 N acts on a body of mass 2 kg. The acceleration is: 1

(a) 20 m/s squared **(b)** 5 m/s squared **(c)** 0.2 m/s squared **(d)** 12 m/s squared

9. The mass of a body on the Moon compared to Earth is: 1

(a) one-sixth **(b)** zero **(c)** the same **(d)** six times

10. The unit of momentum can also be written as: 1

(a) N/m **(b)** N/s **(c)** N m **(d)** N s

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State two ways of reducing friction. 2

12. Define the newton. 2

13. What is momentum? Write its SI unit. 2

14. What is meant by balanced and unbalanced forces? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the momentum of a 2 kg ball moving at 10 m/s and a 0.5 kg ball moving at 30 m/s. Compare. 3

16. Explain the use of seat belts and airbags in terms of Newton's laws. 3

17. Explain why a person standing in a bus tends to fall backward when the bus starts and forward when it stops. 3

OR

State three methods of increasing friction and three of reducing friction. 3

18. A bullet of mass 0.1 kg is fired from a gun of mass 5 kg with a force of 2 N on each. Find the initial accelerations of the bullet and the gun. 3

OR

Name the force of attraction acting on all bodies near the Earth and write the value of g. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A car of mass 1000 kg moving at 20 m/s is brought to rest in 5 s. Find 5
(a) the retardation
(b) the force
(c) the distance moved
(d) the change in momentum.

OR

Explain Galileo's thought experiment with inclined planes and how it led to the law of inertia. 5

20. Explain the third law of motion with five examples and why action-reaction pairs do not cancel. 5

OR

Describe how the motion of a swimmer, a boat and a rocket all depend on Newton's third law. 5

END OF QUESTION PAPER — Total: 40 Marks

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